

FACT-FILLED AND THE MOST HAPPENING CURRENT AFFAIRS FROM AROUND THE WORLD

Robotic arm controlled by thoughts

A new robotic arm is the latest in mind-controlled gadgets that could transform the lives of paralysed and physically challenged people. The technology developed at a US university makes use of a computer connected to a sensor fitted inside the patient's brain. The sensor converts commands from the brain into electrical signals that move the robotic arm.



Cathy, who hasn't been able to move anything below her neck for 15 years, is drinking coffee with a computerised arm that is wired into her

In a demonstration of the robotic arm, a woman who had been paralysed for ten years was able to feed herself a drink using the arm. The doctors discovered that even ten years after someone has been paralysed, the part of the brain that deals with movement continues to work.

Scientists say the new technology is a 'real dream' for people who have lost the use of their limbs. But they say the detailed work isn't finished yet—it still needs to go on for many years before the robotic arm can become available to the masses.

Breakthrough 3D solar cell on the way

Solar3D Inc., based in Santa Barbara, California, USA, is developing a 3-di-

mensional (3D) solar cell technology that will maximise the conversion of sunlight into electricity. The new technology uses a 3D design to trap sunlight inside a photovoltaic structure where photons bounce around until they are converted into electrons. This next-generation solar cell will be more efficient, resulting in a lower cost per watt, making solar power more affordable.

According to Jim Nelson, CEO of Solar 3D, the purpose of develop-

ing the 3D solar cell technology is to achieve greater efficiency and wide-angle light collection, thereby making it possible for the solar cell to produce much more power for the cost.

"We are completing our prototypes and will conduct a pilot run of about 50,000 units next year. Hopefully, we can go to the market by the year end," Nelson added.

Lithium-silicon battery to last five times longer

Researchers at the Stanford National Accelerator Laboratory are developing a lithium-silicon battery that will deliver double and eventually five times the normal battery life.

According to a report, carbon nanotubes have been one of the main reasons

for the incredible advancements in power efficiency. Generally, lithium-ion batteries use graphite anodes to store charge. Only a sixth of the carbon atoms are able to bond with lithium ions, which results in inefficient energy density. If silicon is used instead, four lithium ions can bind with every silicon atom, boosting the energy density to ten times.

However, the only flip side to this technology is that when silicon atoms take in the lithium ions, they swell up to four times in size. When the ions are discharged, silicon shrinks back to its original size. During this process, the entire battery will have to expand and contract. This can lead to a decrease in the battery life or make it completely useless. To deal with this issue, researchers are developing a design that uses a silicon battery anode of double-walled silicon nanotubes, which is coated with silicon oxide.

New technology to turn everything into touchscreen

Researchers at Disney Research and Carnegie Mellon University have created a system that can detect a variety of touch gestures on everyday objects. Touché explores a novel swept-frequency capacitive-sensing technique that can not only detect a touch event but also recognise complex configurations of the human hands and body.

In capacitive sensing, an electrical signal passing through the object changes when touched by a conductive material such as a human finger. Capacitive sensing is already used in smartphone touchscreens, but these devices use electrical signals at only a single frequency. Touché, on the other hand, works with a range of frequencies.

Use of multiple frequencies allows the system to distinguish between a single finger, multiple fingers, a full-hand grasp and many other touch gestures. All you need is a single sensing electrode attached to the object at